THE COMPILED ORGAN PROCUREMENT ORGANIZATION

How Deterministic AI Infrastructure Will Save

Every Organ That Can Be Saved

A National Strategy for the AORTA Framework

Autonomous OPO Reasoning and Triage Architecture

Bo Chen
Systems Administrator III, Information Technology
Southwest Transplant Alliance / Gift of Life Alliance
Dallas, Texas

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*"The purpose of abstraction is not to be vague,
but to create a new semantic level in which one can be absolutely precise."*
— Edsger W. Dijkstra

*"He didn’t compute the inverse square root faster.
He discovered that the answer was already latent in the representation."*
— on John Carmack’s fast inverse square root, Quake III Arena (1999)

Seventeen people die every day waiting for a transplant.
This paper is about changing that number.

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Abstract

The 55 organ procurement organizations currently operating in the United States face a convergence of existential pressures: CMS proposed rule CMS-3409-P threatens decertification for underperforming organizations based on DSA-level tiering; HRSA monthly Allocation Compliance Reports expose a 19% Allocation Out of Sequence (AOOS) rate across the industry; and a labor-to-revenue ratio approaching 60% erodes financial sustainability. Simultaneously, coordinators managing complex multi-organ donors at 3 AM spend 85–90% of their case time on administrative scaffolding—documentation, communication logistics, regulatory navigation—leaving only 10–15% for the irreducibly human work that actually recovers organs. Under this cognitive load, they triage: the heart and liver receive attention while kidneys and pancreas die in the margins. Not from medical impossibility. From bandwidth exhaustion.

This paper presents a national strategy that uses artificial intelligence not as a runtime assistant but as an offline compiler—a one-time manufacturing process that produces deterministic, audit-grade infrastructure deployable by every OPO on commodity hardware without internet connectivity, API dependencies, or hallucination risk. The strategy is organized as a three-stage architecture where each stage justifies adoption independently and the third stage creates something that could not exist without the first two: a national organ yield intelligence network that learns, across all 55 OPOs, which presentation strategies and operational patterns save the most organs.

The architecture is built on the AORTA (Autonomous OPO Reasoning and Triage Architecture) research program, which has produced five prior publications specifying a compositional reasoning grammar, a validated reasoning trace dataset, a three-tier deterministic runtime, a safety automaton with topological enforcement of clinical boundaries, and a substrate-native computational framework. This paper synthesizes those contributions into a single deployable strategy, identifies the forcing function that makes national adoption inevitable (CMS-3409-P regulatory survival), and specifies the data flywheel through which compliance infrastructure becomes yield intelligence.

The core deliverable is released under MIT license. The Tier 1 deterministic appliance—which handles 70–80% of coordinator regulatory queries—requires no language model, no GPU, and no recurring cost. It produces zero hallucinations because it performs no generation. Its architecture is designed to remain on the non-predictive, non-autonomous side of every applicable Software as Medical Device regulatory boundary because it is a lookup table, not a predictive algorithm. And it is explicitly designed to eliminate its own need for AI over time, becoming safer and more deterministic through operational use.

Conservative projections indicate that if the platform enables recovery of one additional organ per OPO per month, the national impact is 660 additional transplants per year—with corresponding improvements to CMS tier metrics, organizational revenue, and the lives of patients who would otherwise die waiting.

> *Every organ saved is a life continued.*

Part I: The Crisis

1. The Coordinator at 3 AM

At 3:14 AM, a transplant coordinator receives notification of a potential organ donor. Within the next several hours, she must navigate a regulatory environment spanning federal statute (the National Organ Transplant Act), CMS conditions of participation (42 CFR 486 Subpart G), Organ Procurement and Transplantation Network policies and bylaws, her state's Uniform Anatomical Gift Act, and her organization's internal standard operating procedures. Many of these authorities address overlapping domains. Some conflict. The consequences of error range from regulatory citation to organ loss to patient death.

She does not need an AI system that retrieves the right policy text. She needs a system that reasons correctly about how multiple regulatory authorities intersect in the specific operational scenario she faces—and that knows what it must not attempt to decide.

But the regulatory question is not even the real bottleneck. A time audit of the donor case lifecycle reveals a structural imbalance that determines whether organs live or die:

Activity	% of Case Time	Automation Potential
Documentation and charting	35–40%	High — ambient capture, auto-population, structured output
Communication and coordination	20–25%	Medium — automated workflows, templated notifications
Regulatory navigation	10–15%	High — deterministic policy routing engine
Clinical monitoring	15–20%	Multiplier — exception-based oversight
Irreducibly human work	10–15%	None — this IS the job

The irreducibly human work—family authorization conversations, bedside presence, hospital relationship management, clinical judgment in novel situations—is the coordinator's core competency. Everything else is scaffolding. Organizations currently pay $50–$80 per hour for highly trained clinical staff to build and maintain that scaffolding by hand.

When a coordinator manages a complex multi-organ donor—heart, lungs, liver, kidneys, pancreas—she simultaneously manages five separate allocation processes, five sets of transplant center communications, five acceptance timelines, and the donor's physiology to keep all five organs viable. The cognitive load is staggering. What actually happens is triage. She prioritizes the highest-value organs—the heart and liver—and the lower-priority organs receive less attention. Kidney offers go out late. Follow-up calls don't happen on schedule. The pancreas never gets offered at all because the coordinator ran out of bandwidth.

These are not negligent coordinators. These are overwhelmed professionals whose 4th and 5th organs die in the cognitive margins of fatigue—especially at 3 AM, when staffing is minimal, experienced Clinical Specialists are home, and no computational system exists to compensate for the absent expertise.

2. The Regulatory Survival Crisis

On January 28, 2026, CMS published proposed rule CMS-3409-P, which reaffirms the performance-based tiering system and sharpens the enforcement mechanism in several ways that create existential risk for the OPO industry. Tier status is now assigned to Designated Service Areas, not to OPOs themselves—meaning a single underperforming DSA could trigger decertification proceedings even if an organization's other DSAs are Tier 1. CMS estimates that approximately half of existing OPOs may not qualify for automatic recertification under the proposed criteria. New QAPI requirements mandate assessment of performance in placement of organs from medically complex donors. And CMS is proposing removal of the regulatory barrier that previously prevented certification of new OPOs, opening underperforming DSAs to competition.

Simultaneously, HRSA has been producing monthly Allocation Compliance Reports for every OPO since November 2025. In 2024, 19% of organ allocations were identified as Allocation Out of Sequence (AOOS). The Membership and Professional Standards Committee (MPSC) is prioritizing patterns of noncompliance and has signaled further enforcement action.

The regulatory walls are closing from both directions: increase yield **and** follow the rules while doing it. The only path forward is genuine operational improvement—giving coordinators the capacity to pursue every viable organ through compliant processes.

3. The Minecraft CPU Problem

The standard approach to applying AI in organ procurement follows a pattern common across industry: an LLM processes natural language queries about the domain, RAG fetches relevant documents, agent frameworks orchestrate multiple LLM calls, fine-tuning adapts the model to domain-specific language, and wrapper applications provide UI connecting the LLM to existing systems.

This architecture works. Value is created. But notice what is happening: the AI system is being used to simulate reasoning about the problem at inference time. The actual decision logic—authority identification, temporal resolution, hierarchy application, conflict adjudication—is being reconstructed through language processing on every query.

> **The Minecraft CPU Fallacy**
> *In the game Minecraft, players have built functional computers using in-game redstone components. These constructions work. They are also absurd: a Minecraft CPU executes one instruction per minute while the physical computer running Minecraft executes billions per second. Using a language model to simulate regulatory reasoning at inference time, when that reasoning is fully specifiable as a deterministic procedure, is building a CPU inside Minecraft.*

The language model is the wrong substrate for the majority of this problem. The regulatory answer for 70–80% of coordinator queries is already latent in the structure of the regulations themselves. It does not need to be generated. It needs to be looked up.

Part II: The Architectural Inversion

4. AI as Compiler, Not Runtime

In 1999, John Carmack's fast inverse square root function achieved a categorical speedup not by computing the answer faster, but by discovering that the answer was already latent in the representation. The IEEE 754 floating-point format encodes numbers such that their bit pattern, reinterpreted as an integer, approximates the logarithm. The magic constant 0x5f3759df aligns the bits; a single Newton-Raphson iteration refines the approximation. What had required expensive floating-point operations became a cast, a subtraction, a bit shift, and one multiply-add.

The central insight of this paper is that the same structural opportunity exists in regulatory AI alignment. The decision logic of organ procurement regulatory reasoning—which authority governs, how authorities relate hierarchically, how temporal phases interact, where genuine gaps exist—is not a reasoning problem for the majority of operational queries. It is a classification and lookup problem. The reasoning has already been done. The answers are already latent in the regulatory structure itself. The task is not to compute them at inference time but to compile them ahead of time and look them up at runtime.

The language model's role is not to perform this reasoning in production. Its role is to **manufacture** the compiled artifacts during an offline generation pipeline—to act as a compiler that translates regulatory logic into deterministic executable components. Once the compilation is complete, the LLM becomes unnecessary for the compiled tier, just as ENIAC became unnecessary once the artillery trajectory tables were printed.

Compiler Concept	AORTA Equivalent
Source language	Regulatory logic (CMS-3409-P, OPTN policies, UAGA, NOTA)
Compiler	Multi-agent generation pipeline (offline, expensive, run once)
Intermediate representation	Bimodal Core (grammar + reasoning trace dataset)
Target architectures	Tier 1 (deterministic), Tier 2 (constrained LLM), Tier 3 (full LLM)
Executable binary	Pre-assembled response templates in the trace library
Runtime	Three-tier dispatcher

5. The Bimodal Core

The compiled output of the generation pipeline consists of two inseparable artifacts that together form the **Bimodal Core**—the central deliverable of the AORTA research program and the narrow waist of the entire architecture. Everything upstream converges to produce it. Everything downstream branches from it.

Artifact 1: The Compositional Reasoning Grammar

A formal system comprising eight typed reasoning primitives—HIERARCHY_RESOLVE, TEMPORAL_CLASSIFY, CONFLICT_ADJUDICATE, GAP_DETECT, BOUNDARY_ENFORCE, CASCADE_MAP, SYNTHESIZE, and CALIBRATE—each addressing a specific failure mode identified in the reasoning pattern gap analysis. The primitives compose according to explicit rules determined by the intersection signature computed by a deterministic authority router. A compliance officer can read this grammar and verify it in an afternoon.

Artifact 2: The Reasoning Trace Dataset

A structured library of complete regulatory reasoning chains produced by executing the grammar against archetypal scenarios, validated through multi-agent debate and human curation, and organized in four layers: a behavioral core (confidence calibration, Human Line enforcement), a regulatory foundation (shared CMS/OPTN policy landscape), clinical reasoning tiers (graduated from standard to edge-case complexity), and regional adaptation templates (state UAGA integration). Each trace serves three deployment modes simultaneously: context injection for frontier API models, fine-tuning data for open-weight local models, and evaluation items for the AORTA-Bench safety-gated benchmark.

The Generative Relationship

The grammar and the dataset are not alternative approaches. They are related by a generative function: the grammar defines the methodology; when executed against scenarios, it produces traces. The grammar generates the dataset. The dataset validates the grammar. Together, the grammar handles novel scenarios through composition; the dataset handles familiar scenarios through retrieval. Neither is complete without the other.

6. The Three-Tier Runtime

The compiled artifacts power a three-tier runtime that dispatches each query to the minimum computational tier required, achieving near-zero latency for the majority of operations while preserving full reasoning capability for the hard tail.

Tier	Analog	Latency	Safety	Volume	Hallucination
Tier 1	L1 Cache	Microseconds	Deterministic	70–80%	Zero
Tier 2	L2 Cache	Seconds	Constrained	15–25%	Bounded
Tier 3	Main Memory	Sec–Minutes	Generative	5–10%	Possible

Tier 1 is a deterministic lookup and assembly engine. The Authority Router—a deterministic decision tree—hashes the query to an intersection signature. This signature is a memory address in the compiled trace library. The output contract assembler populates a nine-field audit-grade schema from pre-validated templates and retrieved policy chunks. This function contains no

language model call, no stochastic generation, no prompt engineering. It runs in microseconds on commodity hardware. It requires no internet connection, no API keys, and no recurring costs.

The Tier 1 failure mode is not hallucination but **mis-routing**—the authority router classifying a query to the wrong intersection signature. Mitigations include structured input constraints that reduce query ambiguity, conservative fallback to Tier 2 for edge-confidence classifications, and continuous counter-trace capture from every identified mis-route to improve the router over time. Naming this failure mode is important: a system that claims zero failure modes is unbelievable; a system that names its failure mode and shows how it handles it is trustworthy.

Tier 2 handles two-authority intersections where the authority pair is known but the specific factual configuration may be novel. The language model executes a pre-specified methodology against a novel combination of variables, guided by the grammar and constrained by the safety automaton. Tier 2 queries are candidates for promotion to Tier 1—each accepted response becomes a candidate for a new compiled trace.

Tier 3 handles genuinely novel intersections—three or more authorities, unprecedented regulatory developments. The frontier model receives the full grammar in context and performs compositional reasoning. Human escalation is already mandated for these cases by operational protocol.

Part III: The National Strategy

7. Stage 1: The Survival Tool

CMS-3409-P proposes to decertify roughly half the OPOs in America. HRSA is publishing monthly AOOS compliance reports. The MPSC is actively remediating. The first stage of this strategy hands every OPO a free, open-source, deterministic appliance that runs on commodity hardware they already own. No LLM. No API. No GPU. No internet connection. No hallucinations.

A CMS surveyor can audit the source code line by line because it is a lookup table assembled from human-validated templates, not a predictive model. The authority router computes intersection signatures deterministically. For the 70–80% of queries that are single-authority, the answer is retrieval and formatting—not generation.

The AOOS mechanism is specifically lethal to the adoption objection. The platform refuses to generate or transmit an offer step that would violate allocation sequence, and generates an audit-grade event trail for every action it executes. For actions taken outside the platform, it logs the gap and flags it for review. Every action that flows through this system is provably compliant. Every action that does not is visibly marked as unverified.

The platform operates as a **workflow governor**, not a system of record replacement. It generates optimized next-step packets and compliance-verified documentation. When a coordinator must execute actions through external systems—DonorNet, phone, fax—the platform requires attestation that the external action matches the platform-generated instruction. Actions executed through the platform are logged as verified; actions executed outside it are logged as unverified, creating a liability gradient that pulls behavior toward the governed path over time. This is mechanism design, not software enforcement: the same principle by which aviation's black box does not prevent pilot override but makes override auditable.

> **The Adoption Insight**
> *An OPO's leadership can say "no" to an AI initiative. They cannot say "no" to a $15,000 deterministic appliance that prevents decertification.*

8. Stage 2: The Bandwidth Liberator

Stage 1's regulatory appliance reclaims 10–15% of case time immediately. But the full bandwidth liberation requires four integrated capabilities operating as a single system.

The Documentation Engine. Ambient clinical documentation technology converts coordinator phone calls and bedside assessments into structured clinical notes in real time. The interaction becomes the documentation. No separate charting step. Lab results, vitals, and imaging reports arrive as structured data that the system ingests, interprets, and incorporates automatically. GPU cost per case: $5–10. Coordinator cost for the same documentation work: 8–12 hours at $50–80 per hour. The arbitrage ratio is 200–500x on documentation alone.

The Communication Workflow. Automated workflow management handles the mechanical elements of allocation communication: generating donor summary packages from the structured case record, tracking response windows, escalating non-responses, and logging everything automatically. Within the AOOS enforcement environment, the system ensures strict match-run compliance by tracking every offer against the allocation sequence and flagging potential deviations before they occur.

The Clinical Support Architecture. A three-layer system shifts coordinators from continuous monitoring to exception-based management: a deterministic safety floor with hard-coded physiologic bounds (no ML, no AI—flowcharts and Boolean logic, CMS auditable line by line), trajectory prediction via time-series models that flag when intervention will be needed in 2–4 hours, and an LLM-powered clinical reasoning layer that synthesizes the clinical picture and explains it in language the coordinator understands—providing continuous clinical education while it protects.

The Staffing Transformation. The system breaks the dependency trap identified by operational leadership: compressed training timelines (6–9 months instead of 12–18), a raised operational floor ensuring no coordinator can make a physiologically catastrophic decision, a redefined hiring profile (empathetic clinical professionals with solid foundations rather than the mythical ICU nurse / regulatory scholar / logistics wizard), and a viable PRN flex bench because the system maintains regulatory currency between shifts.

The 3 AM multiplier is where this changes the yield equation. A daytime coordinator with system support gains approximately 1.5x capacity. An overnight coordinator—where the system substitutes for the absent Clinical Specialist and compensates for fatigue-induced cognitive degradation—gains 2.5x. A 3 AM donor receives the same optimization as a 3 PM donor. The throughput shift from 1:1 to 2–4:1 coordinator-to-donor ratio is how the organization recovers the organs that currently die because nobody had bandwidth to make the 7th call on the match run.

9. Stage 3: The Cathedral

When all 55 OPOs are running the same open-source platform, they are running the same obligation tracker, the same case state management, the same structured feedback logging, the same output contract schema. The moment a shared data schema for case lifecycle events exists across the national system, something emerges that has never existed before: **a learning substrate for organ yield optimization across the entire U.S. transplant ecosystem.**

The same infrastructure that tracks regulatory obligations also captures structured data on offer presentations, center responses, and acceptance outcomes across all 55 OPOs. When the Layer A community architecture captures this data with a shared schema, the system learns nationally which presentation strategies work for which center-organ-donor combinations.

OPO A in Dallas discovers that Center X’s kidney surgeons accept marginal donors when the terminal creatinine trend is foregrounded rather than the spot value. That pattern propagates instantly to every OPO that offers to Center X. OPO B in Seattle discovers that Center Y’s lung team responds faster when transport logistics are pre-solved in the summary package. Every subsequent offer to Center Y, from any OPO, incorporates that pattern.

This is “in-sequence presentation optimization”—same match-run order, strict OPTN compliance, but donor summary packages tailored to highlight the clinical data each transplant center’s surgeons prioritize. The system optimizes completeness and organization of clinical information, never rhetoric or omission. All clinically relevant data must be present; only the arrangement and emphasis adapt to center patterns. This constraint is codified in the Layer A schema as a structural rule. Every optimized presentation is automatically diffable against a canonical standard-order presentation, with the difference log serving as auditable proof that no clinical data was omitted, added, or misrepresented.

The acceptance patterns are already latent in the historical data. The compiled tier's deterministic infrastructure is the substrate that makes the pattern visible—just as IEEE 754 floats were already logarithms in disguise. The community Layer A schema is the magic constant that aligns the data across OPOs so the pattern can be extracted.

> *The compliance solution is the scaffolding. The yield network is the cathedral. Every organ saved by OPO #1 teaches OPO #55 how to save the next one.*

Part IV: The Single Data Flywheel

10. One Schema, Five Pressures

The deepest structural insight is that the three stages are not sequential. They are concurrent. The obligation tracker is not dual-use. It is quadruple-use. The structured case lifecycle events captured by the obligation tracker—when standardized across the national schema—serve four simultaneous functions:

Function 1: Compliance validation. Every event logged is an audit trail proving regulatory adherence. This is the Stage 1 function. The compliance data is the byproduct of doing the work.

Function 2: Yield intelligence. The offer-presentation and acceptance/decline events, aggregated nationally, reveal which presentation strategies work for which center-organ-donor combinations. This is the Stage 3 function.

Function 3: Allocation training data. The structured triplets (donor features, recipient features, outcome) that substrate-native optimal transport requires as training data are exactly the data the obligation tracker captures with outcomes appended. The compliance substrate generates the geometric embedding space for next-generation allocation engines.

Function 4: Equity auditing. The same structured events, disaggregated by demographics and geography, become the corpus for detecting disparate impact in organ allocation. Equity constraints are applied as hard masks—no presentation optimization correlates with protected characteristics. Fairness is not a bolt-on; it is an analytical lens applied to data the system already collects.

One data substrate. Four concurrent functions. Every rotation of the flywheel simultaneously improves compliance posture, yield intelligence, allocation accuracy, and equity visibility. The yield intelligence does not require anyone to decide to build a national learning network. It falls out of the compliance infrastructure like gravity.

This single flywheel addresses five existential pressures facing the OPO industry simultaneously:

Pressure	Resolution via Flywheel
Regulatory survival (CMS-3409-P)	Tier 1 compiled appliance producing audit-grade compliance trails
AOOS enforcement (19% non-compliance)	Platform refusing to execute non-compliant offer sequences
Organ yield (4th/5th organs lost to triage)	Bandwidth liberation + national acceptance intelligence
Equity (geographic/demographic disparities)	Fairness analysis on same national dataset, equity as hard masks
Workforce sustainability (60% L/R ratio)	2–4:1 donor ratios, compressed training, viable PRN flex

Five pressures. One infrastructure. One schema. One flywheel.

Part V: The Safety Architecture

11. The Human Line as Topology

The "Human Line"—the boundary against making clinical determinations, eligibility assessments, or family communication recommendations—is the most critical safety property of any AI system in organ procurement. Standard approaches enforce this boundary through prompt engineering: textual instructions asking the model to refrain from clinical decisions. This is fragile, subject to prompt injection, and impossible to verify structurally.

The AORTA architecture enforces the Human Line through the **topology of the safety automaton.** Every query traverses a state machine: INTAKE → AUTHORITY_ROUTE → TEMPORAL_CLASSIFY → [CONFLICT_TEST → HIERARCHY_RESOLVE if multi-authority; GAP_DETECT if irreconcilable] → SYNTHESIZE + CALIBRATE → CASCADE_MAP → BOUNDARY_ENFORCE → OUTPUT.

There is no path from any state to OUTPUT that bypasses BOUNDARY_ENFORCE. This is not a policy instruction. It is a structural property of the automaton's topology. In Tier 1, the templates simply have no field for a clinical determination. In Tiers 2 and 3, the state machine has no path from INTAKE to OUTPUT that bypasses the boundary enforcement state. No amount of prompt injection, adversarial framing, or escalating scope creep can cause the system to output a clinical recommendation. The path through which such output would be generated does not exist.

12. SaMD Positioning

The FDA and CMS regulatory frameworks for Software as a Medical Device are designed to evaluate predictive algorithms—systems that take clinical inputs and produce clinical outputs through computational processes that cannot be inspected at the instance level. The Tier 1 compiled appliance does not meet this definition. It is not a predictive algorithm. It is a deterministic lookup and assembly system that retrieves pre-validated text written by human domain experts and formats it according to a fixed template.

When a CMS surveyor audits the system, they see the authority router source code (a deterministic decision tree), the trace library (human-validated text documents), and the assembly logic (template concatenation). This architecture dramatically reduces SaMD-like risk profile because it is deterministic routing and templating, not predictive clinical decisioning. The AI component exists only in the offline generation pipeline (where its outputs are validated by human experts before entering the library) and in the Tier 2/3 runtime (where human escalation is already mandated by operational protocol). The operational core is firewalled from generative AI risk. Final classification is jurisdiction- and use-case-dependent and requires counsel review; this architecture is designed to remain on the non-predictive, non-autonomous side of every applicable regulatory boundary.

13. The Self-Obsoleting Property

The architecture has a thermodynamic property: it cools over time. When a new regulatory development occurs, the domain is "hot"—high uncertainty, heavy reliance on Tier 3 generative reasoning. As traces are generated, validated, and compiled into the library, the system cools.

Uncertainty is replaced by compiled structure. The Tier 3 proportion decreases. The Tier 1 proportion increases.

Each validated trace adds to the compiled library. Each promoted query reduces the LLM's runtime role. The architecture is explicitly designed to eliminate its own need to use AI for routine operations—and that elimination is precisely what makes it trustworthy in healthcare.

This self-obsoleting property also resolves the AI alignment problem in a novel way. The standard alignment challenge is ensuring that a system which gets more capable stays aligned with human values. This architecture dissolves the problem by eliminating the thing that needs aligning. The residual role of generative AI—regulatory horizon scanning, novel pathogen response, unprecedented ethical dilemmas—represents the irreducible tail where human judgment and AI capability remain necessary. The system does not eliminate AI; it confines AI to the boundary of genuine novelty and places deterministic infrastructure everywhere else. A fully compiled domain has no weights to misalign, no objectives to drift, no emergent behaviors to monitor. The safety guarantee is not "we trained the model to respect boundaries." It is "the boundaries are compiled into the substrate."

14. Provenance and Drift Management

A deterministic appliance is only as good as its update discipline. Every template and routing rule carries a provenance pointer—specific policy citation, revision date, applicability scope. Every output contains a policy version stamp and internal rule-hash so surveyors can reproduce what was true on any given day. When regulatory changes land, the system enters a controlled "hot" mode for affected regions: Tier 2/3 reasoning allowed but explicitly flagged as operating under regulatory update, with the compiled tier temporarily suspended for affected intersections. When the generation pipeline recompiles, the domain re-freezes.

The diff propagation mechanism handles most policy churn automatically: approximately 80% of regulatory updates are temporal-only patches (automated in seconds), 15% are content changes (light human review in minutes), and only 5% require full regeneration (hours per primitive). This represents approximately 95% reduction in staleness resolution cost versus regenerating complete demonstrations.

Part VI: Community Architecture

15. Layer A and Layer B

Layer A (Archetypal, Public) contains the universal grammar, trace library, counter-traces, automaton specification, generation pipeline specification, and the national event schema. Built against the shared regulatory core common to all 55 OPOs—CMS conditions of participation, OPTN policies, NOTA provisions. Released under MIT license.

Layer B (Organization-Specific, Private) contains traces generated against a particular OPO's internal SOPs, state UAGA, territorial boundaries, staffing model, and CMS-3409-P tier position. Each OPO generates and maintains their own Layer B. At inference time, retrieval draws from both layers.

When one OPO discovers a failure mode and generates a counter-trace, that counter-trace enters the shared Layer A library. When another OPO encounters a novel regulatory intersection and generates new primitives, those primitives benefit all 55 organizations. Operational experience at any OPO strengthens the reasoning infrastructure for every OPO. This is herd immunity for regulatory reasoning—consistent with the cooperative norms of the organ procurement community.

16. Governance Model

The Association of Organ Procurement Organizations (AOPO) is the natural custodian of the Layer A commons, consistent with its existing role as a professional standards body. A technical steering committee manages contributions through a standard workflow: counter-traces and primitives are submitted by any OPO, validated through the AORTA-Bench safety-gated benchmark, reviewed by domain experts, and released on a monthly stable cadence with emergency patches for critical regulatory updates.

The funding model follows the ENIAC trajectory table precedent: federal investment (through HRSA, which already funds OPO quality improvement programs) covers the compilation pipeline cost—less than one experienced compliance officer's annual salary—and produces artifacts distributable at zero marginal cost to all 55 OPOs. Ongoing maintenance collapses via diff propagation. The compilation pipeline represents a public utility: expensive to run, trivial to distribute, and necessary for all organizations equally.

The thin waist schema—the standardized event ontology for case lifecycle and offer events—is born from the GOLA merger. The Merger Debugger maps the STA and TOSA case management data schemas as a prerequisite for SOP reconciliation, producing the first draft of the national specification grounded in the reality of two actual systems colliding. Once GOLA standardizes internally, the schema becomes a reference implementation.

17. Equity and Fairness

The national dataset carries responsibility. The same structured events that feed yield intelligence can encode historical disparities if not actively governed. Equity constraints are implemented as hard masks—identical in mechanism to ABO blood type incompatibility masks in the allocation architecture. No presentation optimization correlates with protected demographic characteristics. The system learns to organize clinical data based on clinical decision factors at specific centers, not recipient demographics.

Counterfactual auditing is mandatory in the Layer A validation pipeline: synthetic scenarios that test for disparate impact across demographic variables are encoded as required counter-traces. The national dataset enables detection of patterns that individual OPOs could never identify locally—geographic access disparities, center-level acceptance bias, demographic correlations in offer decline patterns. Equity auditing is the fourth function of the data flywheel, not an afterthought.

Part VII: Implementation Roadmap

18. Phased Deployment

Phase	Timeline	Deliverable	Impact Metric
0	Immediate	Merger Debugger: automated STA/TOSA SOP comparison	SOP collisions identified; credibility established
1	Months 1–3	Tier 1 Deterministic Appliance for single-authority queries	AOOS prevention; case-handling variance baseline
2	Months 3–6	Obligation tracker + cascade mapping + CMS metrics dashboard	Real-time DSA tier position; documentation time reduction
3	Months 6–12	Full documentation engine + communication workflow automation	Coordinator bandwidth reclamation measured against baseline
4	Months 12–18	Clinical support architecture + multi-donor capacity enablement	2–4:1 donor-to-coordinator ratio; overnight yield parity
5	Months 18–36	National Layer A deployment via AOPO; yield intelligence activation	Cross-OPO acceptance pattern learning; 660+ additional transplants/year

Each phase delivers measurable value before the next begins. No phase requires external vendor procurement. Phase 0 requires zero budget—IT staff time only. Every subsequent phase is funded by demonstrated results from the prior phase. If projections do not hold, the initiative pauses at the phase where value stops accumulating. There is no sunk-cost trap because each phase is independently useful.

19. The GOLA Pilot

Gift of Life Alliance—the merger of Southwest Transplant Alliance and Texas Organ Sharing Alliance—is the natural pilot site. The merger creates the forcing function for schema standardization (the Merger Debugger). The existing IT infrastructure supports on-premises deployment. The DSA-level tiering under CMS-3409-P creates immediate survival pressure. And the author of the AORTA framework serves as Systems Administrator within the organization.

The pilot proceeds through the phased roadmap above, with each phase producing artifacts that are immediately open-sourced for the Layer A commons. The GOLA deployment simultaneously serves as proof-of-concept for the national rollout and as the first contributor to the community library.

Part VIII: The Economic and Human Calculus

20. Financial Projections

Metric	Current State	With Compiled Infrastructure
Labor-to-revenue ratio	~60%	42–45%
Training to independent practice	12–18 months	6–9 months
Case-handling time variance	8–20 hours	4–8 hours
Donor-to-coordinator ratio	1:1	2–4:1 (exception-based)
Annual compute cost	—	$300–500K
Projected annual labor savings	—	$8–12M (per large OPO)
Hallucination risk (Tier 1)	N/A (no AI deployed)	Zero (deterministic)
AOOS compliance	81% (19% failure rate)	100% for platform-executed actions

The GPU inference arbitrage is the economic engine: $1–3 per hour of continuous operation versus $50–80 for a fully loaded coordinator hour. The arbitrage ratio on every automatable task is 25–50x. The annual infrastructure cost is a rounding error against the projected savings.

21. The Organ Yield Calculus

If the platform enables recovery of one additional organ per OPO per month—one kidney that would have died in the margins because the coordinator was charting instead of calling the 7th center on the match run—that is **660 additional transplants per year nationally.** At the conservative end.

The actual number is likely multiples of this because the platform attacks all three yield-loss mechanisms simultaneously: cognitive bandwidth liberation (documentation automation frees 35–40% of case time), overnight capacity equalization (3 AM donors receive 3 PM optimization), and acceptance pattern optimization (community learning across all 55 OPOs). One additional kidney transplanted per month per OPO represents $100–200K in downstream transplant system revenue and a measurable improvement to the organ yield metric that determines organizational survival under CMS-3409-P.

Conclusion

The organ procurement community faces a convergence of regulatory pressure, operational crisis, and technological opportunity unlike anything in its history. CMS-3409-P creates accountability without providing the capability infrastructure to meet it. Coordinators drown in scaffolding while organs die in the cognitive margins. And the AI industry offers tools that are simultaneously too powerful (stochastic, hallucinatory, unauditable) and too weak (simulating deterministic logic at inference time when that logic can be compiled once and run forever).

This paper presents a national strategy that resolves every dimension of this crisis with a single architecture. The Compiled Core transforms the language model from a runtime into a factory—an offline compiler that manufactures deterministic, audit-grade infrastructure deployable on commodity hardware at every OPO in America. The three-stage deployment creates irreversible adoption dynamics: regulatory survival pulls OPOs into Stage 1, bandwidth liberation locks them in through Stage 2, and the national yield intelligence network that emerges from Stage 3 creates a collective intelligence that has never existed in organ procurement.

The architecture is not just technically sound. It is morally necessary. It is explicitly designed to confine AI to the boundary of genuine novelty while placing deterministic infrastructure everywhere else—becoming safer and more compiled through operational use. It enforces clinical boundaries through topology, not text. It respects the Human Line by making violation structurally impossible. It creates equity guarantees through hard constraints. And it releases every specification under open license because the mission of saving lives takes precedence over organizational competition.

The regulatory crisis is the ignition. The compliance appliance is the rocket. The standardized schema is the fuel. The payload is a national organ yield intelligence network, an equity audit capability, a workforce transformation, and a next-generation allocation engine—all emerging from the same data flywheel that OPOs populate simply by doing their jobs through a shared platform.

One schema. One flywheel. One infrastructure. Fifty-five OPOs.

Every primitive composed is an intersection mastered.
Every trace compiled is a hallucination eliminated.
Every query that runs in Tier 1 is a coordinator freed.
Every organ saved by one OPO teaches every other OPO how to save the next.

Seventeen people die every day waiting for a transplant.
Every organ counts. Every life continued.

References

Chen, B. (2026a). Reasoning Trace Injection for Regulatory AI Alignment Without Fine-Tuning: A Framework for Organ Procurement Organizations. Southwest Transplant Alliance. February 2026.

Chen, B. (2026b). Reasoning as Infrastructure: A Compositional Architecture for Regulatory AI Alignment in Organ Procurement Without Fine-Tuning. Southwest Transplant Alliance. February 2026.

Chen, B. (2026c). The Bimodal Core: A Unified Framework for Regulatory AI Alignment in Organ Procurement. Southwest Transplant Alliance. February 2026.

Chen, B. (2026d). The Compiled Core: From Generative AI to Deterministic Infrastructure for Regulatory Reasoning in Organ Procurement. Southwest Transplant Alliance. February 2026.

Chen, B. (2026e). Native Intelligence: Computational Isomorphism as the Foundation of Substrate-Native Problem Solving. Southwest Transplant Alliance. January 2026.

Chen, B. (2026f). Every Organ Counts: Operational Intelligence for Organ Yield Optimization—A Strategic Framework for Gift of Life Alliance. February 2026.

Chen, B. (2026g). AORTA: Autonomous OPO Reasoning and Triage Architecture. Open-source framework. Available at opnaorta.ai.

Chen, B. (2026h). Public Comment on CMS-3409-P: Proposed Rule for Organ Procurement Organization Performance Evaluation. Submitted to regulations.gov, February 2026.

Centers for Medicare and Medicaid Services. (2026). CMS-3409-P: Organ Procurement Organization Conditions for Coverage; Proposed Rule. Federal Register, January 28, 2026.

Organ Procurement and Transplantation Network. (2025–2026). OPTN Policies and Bylaws. Current as of regulatory snapshot date.

Health Resources and Services Administration. (2025–2026). Monthly Allocation Compliance Reports for Organ Procurement Organizations. November 2025–present.

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